



# An overview of land use conflicts in mining communities

Gavin Hilson

*Environmental Policy and Management Group, Imperial College Centre for Environmental Technology, Royal School of Mines,  
Prince Consort Road, London SW7 2BP, UK*

Received 14 March 2001; received in revised form 22 August 2001; accepted 28 August 2001

---

## Abstract

This article examines the causes and impacts of land use conflicts between large-scale mines and community groups, and identifies a series of (land use) conflict resolution strategies for mine management. It is becoming increasingly challenging for mines, which demand a significant amount of area to operate, to coexist with the indigenous people of surrounding communities who depend largely upon the land for their livelihoods. The article contends, however, that most of the unavoidable environmental problems that occur at sites—namely erosion, sedimentation and vegetation removal—are largely dismissed by locals, but that poor communications and highly preventable environmental accidents have been, and continue to be, the chief causes of intense land use disputes between mines and surrounding communities. As governmental intervention is often miniscule, most of the responsibility rests with mine management to ensure that land use conflicts are effectively prevented and resolved. Whilst no de facto strategy exists that will completely satisfy both parties, quite conceivably, compromises can be reached if: community consultation between the parties is significantly improved; regional governments assume a leadership role in coordinating the efforts of international agencies; appropriate compensation packages are provided to impacted communities; and partnerships are forged between large- and small-scale miners. © 2002 Elsevier Science Ltd. All rights reserved.

**Keywords:** Mines; Mining; Land use; Conflicts; Disputes; Ok Tedi Mine

---

## Introduction

Perhaps no single industry has precipitated more disputes over land use than mining. Though economically, a great number of rewards are reaped from its activities, the land demands placed by mines often cause severe community disruption and hinder the development of other potentially profitable industries such as small businesses, merchant services and small-scale fisheries. Furthermore, intensive mining activity can cause major environmental complications, which in turn can render land unsuitable for a number of other important industrial applications—namely farming and forestry—after reclamation.

Disputes over land often occur between mine management and community groups. As many of these conflicts are unforeseeable from the outset, few mining companies have in place appropriate land use policies for conflict resolution. Circumstances are severest in developing countries, where governmental intervention is minimal, regulatory frameworks are commonly incom-

plete, and fewer effective support schemes are in place for community and industrial groups. Compounding the problem is the fact that a number of these countries' governments, which have heavily promoted foreign investment in their minerals sectors in recent years, almost exclusively side with mining companies on key land use issues. Clearly, without the full cooperation of governments, most of the responsibility rests with mining companies to resolve land use disputes of all types, and to ensure that mechanisms are in place to prevent further community disruption.

The purpose of this paper is to examine the causes and impacts of land use conflicts between large-scale mines and community groups, and to identify a series of (land use) conflict resolution strategies for mine management. In the first of three sections that follow, a detailed overview of land use disputes in mining communities is provided in which causes and impacts are examined. This is followed by a case study of the Ok Tedi Mine in Papua New Guinea, which effectively illustrates how improper coordination and neglect on the part of mine management can induce a series of land use conflicts in surrounding communities. The paper concludes with a

---

*E-mail address:* g.hilson@ic.ac.uk (G. Hilson).

discussion that describes how a series of strategies—namely, improved community consultation, governmental coordination of research activities, the provision of appropriate compensation packages to impacted communities, and the formation of working partnerships between large- and small-scale miners—could aid in the prevention and reduction of land use disputes between mining companies and indigenous people.

### **Land use conflicts in mining communities: causes and impacts**

As Castro and Nielsen (2001) explain, natural resource conflicts are typically “severe and debilitating, resulting in violence, resource degradation, the undermining of livelihoods, and the uprooting of communities”, and if not addressed, “can threaten to unravel the entire fabric of society”. In short, “each party wants to pursue its own interests to the full, and in doing so ends up contradicting, compromising, or even defeating the interest of the other” (Ochieng Odhiambo, 2000). In the context of mineral extraction and processing, activities can be extremely environmentally destructive and over the course of many years, can cause irreversible damage to surrounding landscapes. Of these, a great number are unavoidable, and can only be remedied at the time of mine decommissioning. For example, large tracts of forest are often removed to gain access to underlying ore bodies, and a vast quantity of often-productive topsoil, through continuous exposure to weathering agents, erodes and collects in nearby water bodies. However, communities are well acquainted with and often prepared for these complications; it is the series of unexpected, often preventable, land contamination problems—namely chemical spillage, collapses of tailings dams and heavy metals leaching—that cause the greatest disruption and in turn induce disputes. Poor communication, broken promises and a lack of working agreements between mines and indigenous peoples further intensify these conflicts. This section of the paper seeks to provide an overview of the causes and impacts of land use disputes between large-scale mines and community groups.

#### *Impact of mining on communities*

Where communities are forced to bear the full brunt of mining impacts, they cannot be assumed from the outset to be inherently in favour of local mineral development. In fact, the indigenous groups that have occupied a region for thousands of years typically view a 20 or 30 yr mining project as “a disruption in the flow of time, much like the effects of an earthquake or major flood” but having “the potential to be much more disruptive to their way of life” (Asp, 2000). Their

attitude towards a local mine depends largely upon local circumstances and factors including income, the degree of unemployment, land use and land ownership institutions, perceptions, and political and social values. Thus, the way in which a mine manages its relationship with the surrounding community is a key determinant in its long-term success.

It is a well-known fact that mines can have a positive socioeconomic impact in local communities. Companies often contribute to the development of key socioeconomic infrastructure such as roads, hospitals, schools and housing. Moreover, the revenues accrued from activity contribute positively to export and foreign exchange earnings, and at the community level, projects serve as a major source of employment for local people, and trigger the rise of a wide range of small businesses such as catering, transport and cleaning services. In Canada, for example, mining accounts for approximately 15% of national exports and 4–5% of national GDP, and provides highly paid jobs to more than 340,000 Canadians in 150 communities (Hilson, 2000a). In another example, in Zambia mining activities contribute 20% to GDP, account for 90% of export earnings, and employ 15% of the country’s workforce (MBendi, 1999). Perhaps a more significant issue from the standpoint of employment is the fact that mining operations, because of a propensity to be located in rural areas, commonly induce major changes in local demographics, precipitating migration from neighbouring villages and townships. A large percentage of those that migrate are both redundant mine labourers relocating for the purpose of securing employment at the mine, and small-scale mining entrepreneurs seeking to exploit portions of the ore body being worked. In the Northern Brazilian State of Pará, for example, 400,000 individuals have come to the area seeking riches by tapping into portions of the resident mine’s resources, 100,000 of whom now live directly beside the gold mine (Lemieux, 1997). In cases like these, the mine represents the sole source of income, or is the major industry in a region or community.

However, despite bringing a wealth of socioeconomic benefits, mines can also cause a number of adverse impacts in communities. Basic examples include, inter alia, the following:

- a loss of ability to hunt, fish and gather,
- a loss of freedom of movement,
- locals being forced to resettle or relocate,
- a fundamental disrespect for traditions.

The most significant disruptions caused by large-scale mining activity result from major demographic changes. Although sites provide numerous indigenous peoples with higher paying jobs, as Sweeting and Clark (2000) explain, major population influxes—namely, the migration of thousands of foreign and non-local employees to

the area occupied by a mine—can upset the social balance of local communities, impacting local water supplies, resource availability and regional hunting. Perhaps the biggest concern with mass resettlement is the spread of diseases such as yellow fever and tuberculosis. One notable example was in Venezuela, where, between 1991 and 1995, 25% of the Yanomami people died from malaria introduced to their community by migrating artisanal gold miners seeking to exploit portions of large-scale mineral deposits (Vital Statistics: Mining and Canada, 1997). The demographic shifts caused by mining can also lead to increased prices for local goods and a greater widening of disparities among local populations. Such was the case in the Peruvian town of Cajomerca, located 48 km from Yanococha, South America's largest gold mine. Although the community, which has a population of 180,000, benefited from the creation of 1200 new jobs and US\$7 million in roads, schools, and local infrastructure, at the same time, housing prices, education and staple goods increased in price as a result of the presence of higher paid (mine) employees. In fact, in some parts of the town, living costs have increased tenfold (Dorian, 1996; Sweeting and Clark, 2000).

In short, a mine can have an adverse impact on local resources and traditions, and the demographic changes induced by its existence can cause significant social disruption in local communities.

### *Industrial disputes*

Before examining more in depth the direct impacts mine land use has on communities and indigenous peoples, it is important to first briefly note that initially, land use disputes can occur between mining companies and other industries. It has been primarily forestry and agricultural operations that have competed with mining companies for land, since both require massive surface area to flourish. Perhaps the most well documented case is Brazil, where there is a long history of competition between local miners, and farmers and ranchers over land. Here, friction is particularly intense between migrating *garimpeiro* gold miners and small-scale cattle herders and rubber planters. Furthermore, as Pedlowski et al. (1997) explain, the use of mercury—a toxic leach reagent used in *garimpeiro* mineral processing activities—has caused irreversible damages to vast, potentially productive, regions of the Brazilian Amazon, and with the bulk of these operations growing and being conducted outside the law, it is becoming increasingly challenging to assess their scale and intensity.

However, conforming more to the objectives of this paper, one of the most often-overlooked impact large-scale mines have on communities is the displacement of local small-scale miners, most of whom are either redundant large-scale mine workers or local villagers.

This segment of the industry features the most rudimentary of techniques and technologies, and produces some one-sixth of global non-fuel output (Hilson, 2001a). Today, between 11.5 and 13 million people are employed at small-scale mines, although it is estimated that as many as 30 million depend on the incomes generated in the industry both directly and indirectly (ILO, 1999). In recent years, a number of developing countries (e.g. Ghana, Zimbabwe, and the Philippines), in an attempt to create a *de jure* small-scale mining sector, have introduced licensing systems whereby individual small miners are required to register by law and obtain a permit for engaging in mineral extraction and processing activity on a given plot of land. Most of these systems are unfortunately unsophisticated, and quite often, small-scale miners are granted legal authority to mine minerals on already-existing large-scale mining plots; this in turn precipitates land use disputes between the two parties.

Ghana is one country that is characterized by a long history of clashes between small miners and (large-scale) mine management over land. Here, land use conflicts have mainly occurred within the Tarkwa and Ashanti (near Oda) Regions of the country—more specifically, in close proximity to the country's richest gold deposits. Illegal *galamsey* commonly mine within the concessions of large-scale mining companies, and because of their cultural relationships with the land—namely, ancestral ties—a significant number are reluctant to relocate. For example, in 1990, seven cooperative groups of small-scale miners had been registered in an area of 155 acre within the same concession awarded to the Ghanaian mining company Abosso Goldfields Ltd. (AGL). Some 600 small miners, many of whom claimed to be amongst the first occupants of the region, encroached on the exploration trench areas of AGL. It became extremely challenging and laborious for AGL to chase small gold miners off “their” exploration area, and after the army and police were called upon to intervene, violent clashes erupted (Agyapong, 1998). Similar conflicts occurred at the neighbouring site of Teberebie Goldfields Ltd. in 1991. Several *galamsey* are still unwilling to leave the area, contending that they have no alternative “source” of livelihood, and although their activities have caused significant environmental damages in various portions of the concession, they refuse to take any responsibility for these damages, and have not taken any necessary safety precautions. Complications have even occurred at Ashanti Goldfield's Obuasi property, Ghana's largest, state-of-the-art gold mine. Violent clashes erupted in 1996 between *galamsey* and local security forces, which resulted in over US\$1 million in damages at the site, and more recently, in December 2000, vandalizing *galamsey* set ablaze the poultry farm at Obuasi and stole valuable livestock. This issue is particularly pressing and in need of immediate attention because productive land is

disappearing at an unprecedented rate throughout the country: some 1.05% of cropland and 1.26% of forest cover is lost each year (World Bank, 1998) to desertification, industrialization and urbanization.

### *Mines, communities, and land use conflicts*

As already indicated, most mines are situated within rural areas—mainly in locations of previously uncleared forest, or where people make use of land for agricultural and subsistence purposes. Though the industry typically supports the view that with good technology and managerial skills, there should be no reason for preventing the opening of any mine, it is important to clarify that in even the most advanced of living conditions, above all, soil or land will always remain the medium that satisfies primary human needs for food and shelter (Verheye, 1997). Conflicts, therefore, inevitably result between communities and mines simply because both place fundamentally different socioeconomic values on land.

The most significant issue often overlooked by mine management is the actual cultural impact operations have on Aboriginal and indigenous people. As Epps and Brett (2000) explain, these groups are commonly amongst the poorest segments of the population, and often engage in local economic activities such as small-scale agriculture, forestry and fisheries. Because groups typically reside in small, simple communities that are largely unexposed to global society, they are particularly vulnerable to the negative impacts of development, making it critical that important cultural resources are best managed. The strategy of a number of mining companies in the past was to provide both financial compensation and social support in exchange for land. However, because of strong ties to the environment dating back thousands of years, monetary compensation rarely makes up for land loss. For groups such as the Iroquois and Huron Indians, which have resided in certain parts of North America for over 20,000 years, and the Aborigines of Australia, who are arguably the first humans to occupy Oceania, it is unrealistic to assume that any sum of money could compensate entirely for the disruptions in lifestyle caused by local mining activity. Their main concerns when it comes to the issue of land use policy can be summarized as follows (modified from Aguilar, 2001):

- Self-determination, including demands related to property rights over land and resources;
- prior informed consent in relation to the protection of traditional knowledge and plant medicines;
- cultural rights that include the right to express and maintain different cultures, the right to religion, language, and access to sacred sites and religious practices;

- rights to communal property in lands and territories; and
- right to control the traditional knowledge of property

It is therefore not surprising that relationships between mine management and groups like these are typically fragile, and because mines are long-term projects, often, when a failed commitment results, conflict ensues.

In principle, the mine requires a given area to excavate, but unplanned expansion and environmental accidents lead to further displacement of land. For example, chemical leakage, Acid Mine Drainage (AMD), and breaches in waste storage facilities can contaminate surrounding soils, and threaten local fisheries and agricultural activities. Further, waste tailings from mines, which contain elevated levels of heavy metals and chemical residues, threaten surrounding farmland and impose a major threat to human health. Ensuring their safe containment—more specifically, constructing a tailings disposal facility—also requires use of additional land that potentially plays an important socioeconomic value in a community.

Mining land use conflicts are typically most intense in the developing world, where the issue of land tenure—more specifically, clarification of who actually *owns* land—is the cause of most problems. In many developing countries, residents have been awarded legal title to large percentages of land. For example, as McLeod (2000) explains, customary tenure in Fiji, where indigenous landowners believe they “own everything below and above...including minerals”, is 84% of all land. In other case, in New Guinea, over 90% of land is owned by residents (MiningWatch, 2000). Therefore, when it comes to negotiations about prospective mining projects, input from these individuals is most important. However, there is the tendency that in countries where communities have the most “say” in mining projects, governmental support is weak, in which case, residents are poorly informed about key mining issues and are hence easily persuaded in mining matters. This compounds problems in the event of mining complications since these governments, which perceive mines as significant contributors to national income, almost exclusively side with mine management and rarely come to the aid of impacted communities. Furthermore, in most countries, governments retain the rights to sub-surface minerals, and can therefore grant rights to a large-scale mining company seeking to explore and extract minerals even on land that has been legally awarded to another group (Sweeting and Clark, 2000).

Occasionally, people are displaced outright and are forced to settle in other rural regions. As already noted, the typical strategy of the mining company is to provide monetary support to communities, and in the event of relocation, to finance the development of housing in an uninhabited area and to ensure that living conditions are

improved overall as a result of the relocation. One case in which successful relocation occurred, and where land use conflicts were avoided was at Goldfields' Ghana's Tarkwa mine. As Steyn and Kahle (1998) explain, the lives of some 20,000 people had been disrupted as a result of its operation. In 1996, negotiations began with surrounding communities and a resettlement programme was implemented. Settlers received upgraded housing from previous wattle and daub structures; houses that contained a utility room for bathing, a significant upgrade from previous non-existent sanitation facilities; compensation for crops, including consideration for land tenure; (if owning at least one bedroom) received 150% of its market value, in addition to an allowance of US\$1000; and access to a community with storm water drainage, two schools and a marketplace. In the end, with minimal dispute, the mine was able to gain access to land containing minerals, and indigenous people, as a result of relocating, improved their quality-of-life.

However, not all communications and agreements between mining companies and community groups over land have been harmonious. One such case is the Ok Tedi Mine in Papua New Guinea, which illustrates effectively how irresponsible mining activity can cause major disputes between local villagers and mine personnel. The case of Ok Tedi is examined more in detail in the next section of this paper.

### A case study of Ok Tedi Mine, Papua New Guinea

Papua New Guinea is a mountainous, tropical island, where the great majority of people reside in remote villages and relies on the surrounding land for survival. It is therefore not surprising that mining activity, which has intensified throughout the island in recent decades, has had wide-ranging impacts. The adverse impacts of large-scale mining in Papua New Guinea have been well documented. In addition to causing significant damages to landscapes, and contaminating soils used for subsistence agricultural purposes, intense activity has led to the displacement of communities, the spread of disease, and widespread poverty. With the exception of perhaps the Bougainville site, no mine has had as negative an impact on the people of Papua New Guinea as the Ok Tedi mine.

The Australian mining company BHP Minerals began prospecting at the Ok Tedi site in the 1970s and began excavation during the mid-1980s. It is now one of the biggest open-cut copper and gold mines in the world. As Low and Gleeson (1998) report, the project has had enormous impacts on the national economy. Outputs from the mine account for at least 16% of annual export earnings and 10% of national GDP, with annual after-tax profits of approximately US\$200 million. Yet, in

spite of these contributions, which have helped to eradicate much of Papua New Guinea's national debt, the mine has caused, and continues to cause, significant damages to the landscape. More specifically, it has destroyed over 1000 km<sup>2</sup> of virgin rainforest and wetlands, has gouged an enormous crater in the mountain, and each day discharges some 80,000 tonnes of limestone sludge containing copper and toxic chemical residues into the upper regions of the Ok Tedi River, causing it to rise 4–5 m in some places. Swales et al. (2000) confirm that these chemical wastes, along with destruction of habitat from mine sediment, have been responsible for significant declines in fish populations. It is estimated that over 250 million tonnes of waste have been released into the surrounding environment since the mid-1980s, which has destroyed arable land and thousands of sago trees; the total area of dead forest now exceeds 2700 km<sup>2</sup>.

Careless mining environmental management has caused additional damage to surrounding landscapes and water bodies. For example, as Harper and Israel (1999) report, wide areas of land adjacent to a copper slurry line were contaminated when it fractured and released slurry. Furthermore, in June 1984, a barge carrying containers of sodium cyanide and hydrogen peroxide lost its cargo, setting free 180 drums into the Ok Tedi River, and although many were recovered, others, for a number of economic reasons, were not. It is estimated that some 140,000 of cyanide were lost to the river, which, in combination with the 1000 m of untreated tailings discharged later that month, caused mass fish, prawn, turtle and crocodile deaths.

Most significantly, intensive mining at Ok Tedi has led to the displacement of a number of native *Wopkaimin* communities. Their culturally diverse way of life was significantly disrupted when the mine began operating in the mid-1980s; more than 40,000 *Wopkaimin* were sustained by the fish, forests and plains of the Fly River system. They are the original inhabitants of the area, and consider Mt Fubilan, the site of rich mineral deposits and the location of the Ok Tedi mine, as sacred. Their first encounter with the outside world was in 1953, and before this exposure, there was a strong gender division in tribal culture, particularly with respect to food acquisition and consumption (Mckillop and Brown, 1999). Fortunately, the majority of the *Wopkaimin* initially lived away from the first prospectors that arrived in the 1960s and were thus able to retain traditional customs, and maintain the subsistence production autonomy in their ancestral rainforests (Hyndaman, 1997). However, when additional prospectors invaded in the 1970s, the *Wopkaimin* reluctantly chose to surrender small areas of land in exchange for small sums of money. They later agreed to lease more land—representing a total of 20% of local hunting

grounds—in exchange for additional compensation, a share of royalties, offers of employment, and the establishment of educational, social and medical facilities (Harper and Israel, 1999).

Land use conflicts between the *Wopkaimin* and the management of Ok Tedi began in the mid-1980s. Groups lobbied for BHP to compensate families for damages to land, and approached the national government for support on numerous occasions. As one local landowner claims, “with the mining operation...life has changed...resulting in loss of rainforest, and degradation of the local environment. Rural village subsistence farmers depend on the environment for survival” (Hettler and Lehmann, 1995). Another claims that their lives “...depend on the river system” and that because “it has changed colour, from clean to milky colour...food is being lost” (Harper and Israel, 1999). The mine has also caused the river to flood, which in turn has displaced vast areas of cultivated lands and thousands of trees, and the once fertile soils that hundreds of villagers rely on for growing staple foods have been coated with thick grey sediment (tailings). In many places, following periods of heavy rains, trees, swamps and creeks in close proximity to riverbanks have been completely covered (King, 1997).

Unable to reach an adequate compensatory agreement with BHP, plaintiffs Rex Dagi and Alex Main, both from the indigenous Yonggom villages along the Ok Tedi, travelled to Holland, Germany, Brazil and the US to protest the mine’s impact and to meet with international conservation agencies. The original compensatory agreement stipulated providing US\$6.6 million in compensation to villagers in the form of meeting halls, fresh water and shower blocks. However, this did not include villagers downstream, who lobbied for some US\$2 billion in compensation (Low and Gleeson, 1998; Harper and Israel, 1999). Though a package of US\$56 million was promised to villagers, the case went to court, but in 1996, an out-of-court settlement was finally reached, under which BHP would stop dispensing tailings into the Ok Tedi/Fly River system; US\$56 million in compensation would be provided to villagers; and an additional US\$20 million would be provided to downstream villagers most affected by waste effluent (Low and Gleeson, 1998). An additional 10% equity share in the mine was also awarded to the province (Western Province) in which it operates.

The Ok Tedi case, however, is not unique. Land use disputes of this nature have occurred throughout the world. The common factor in these and related cases is that promises have been unfulfilled, and mining environmental impacts to land have not been properly prevented. The next and concluding section of this paper prescribes a series of strategies for reducing the intensity of mine–community land use conflicts.

## Recommendations and conclusions

It is unrealistic to assume that agreements can be reached in which all parties will be *completely* satisfied. Given that mines, communities, and industries alike compete for limited plots of land, it is highly unlikely that strategies can be devised that provide each with a maximum benefit. It is therefore not the intention of this discussion to argue that superlative strategies are possible in such cases; rather, it is suggested that with improved strategizing, satisfactory compromises *can* be reached, in which both the mine and the community can coexist in harmony. With a number of governments providing very little in the way of assistance, most of the responsibility rests with mine management to ensure that positive relations with communities are maintained.

The first recommended action is that mines improve communications with communities. As Hilson and Murck (2000) explain, effective communications are essential in an industry like mining, and that it is crucial, for the benefit of surrounding communities, that public meetings be held in which locals can express concerns, provide input to various development phases of the mine, and have questions answered by mine personnel. In fact, regular communication should be maintained in key decision-making processes during the extraction, processing, excavation, construction and abandonment phases. Some community consultation techniques that can be deployed by mine management include, *inter alia*, the following (adopted from EPA, 1995):

- Arranged visits to mines—the aim of which would be to show residents how the mine is making use of the land, and what environmental protection measures are being used to prevent contamination.
- Attitudinal surveys—the process of obtaining baseline information from communities on key land impact studies.
- Partnership links—the process of developing formal links between mineral operations and local schools, which can be important in helping a community gain better understanding of land use issues.
- Liaison groups—forming groups that specifically interact with the community and provide direct feedback to mine management.
- Public meetings—perhaps the most obvious and useful of strategies, whereby communities are informed about a formal gathering with representatives from government, the mine, and local environmental agencies, and where residents can voice concerns.

Although the measures proposed appear generic and vague in scope, their effectiveness as community consultation strategies was confirmed in April 2000 in Ottawa, Canada, where 50 people from eleven countries gathered to share stories about “the devastation large-scale mining has brought on people’s lives, the lives of

their communities, and the land itself” (MiningWatch, 2000). The group quintessentially reached a consensus that the aforementioned methods, along with local capacity building strategies and the building of a global community-based network, are keys to alleviating conflicts between mining companies and indigenous peoples over land.

A second recommendation is for regional governments to assume some responsibility in coordinating the efforts of local mining companies and international agencies in the area of mining land use policy. As already noted, the governments of a number of developing countries such as the Philippines, Ghana, Argentina and Mexico give preferential treatment to large-scale mining operations because they make important contributions to regional foreign exchange and export earnings. Needless to say, these governments are still keen on resolving regional mine–community land use disputes, their actions merely underscoring how the developing world, most of which is highly in debt, is very much at the mercy of large-scale mining activity because of their potential economic contributions. In recent years, however, important strides have been made internationally to address mining–community land use disputes and related problems. The launching of the Global Mining Initiative (GMI) signifies an important step toward improved relations in this area. As Hilson (2001b) explains, the purpose of the Initiative, which brings together many of the world’s largest mining and mineral companies, is to produce a thorough analysis of the industry’s important sustainable development issues. Commissioned through the World Business Council for Sustainable Development (WBCSD) and designed by the London-based International Institute for Environment and Development (IIED), the GMI will cover a broad range of environmental and socioeconomic issues, including, *inter alia*, waste management, human rights, biological diversity and corporate governance. A global conference on mining and sustainable development will be held in September 2002, at which the significant findings of the research will be presented. In the meantime, however, it is recommended that governments forge partnerships with international institutions with vested interest in mining/community development issues. The World Bank, for example, has a mining research group (“World Bank Group Mining”) that undertakes research in a wide range of areas including community development, closure, sector reform and AIDS. It also provides loans and grants to governments keen on financing the development and implementation of policies that address the aforementioned issues, and has already carried out important projects in Ghana, Ecuador and Argentina. Another prominent think-tank, US-based Resources for the Future, undertakes both land use and mining research in developing countries. Recent undertakings include land spatial management,

resource productivity growth and environmental management. There is also merit in forging partnerships with more localized non-profits. The South Pacific Applied Geoscience Commission (SOPAC), for example, which undertakes research in Australia and neighbouring South Pacific islands, seeks to improve the well being of Pacific Island member countries through the application of geoscience to the management and sustainable development of their non-living resources (SOPAC, 2001). In another example, the Chilean government has established the Empresa Nacional de Minería in an attempt to promote mining activities and improve the sustainability of operations (Hilson, 2000b). In these and related cases, there is opportunity to target specific problems as opposed to incorporating them into a broader agenda of issues, the typical approach of larger international non-profits.

It is equally imperative that mine management provides appropriate compensatory packages and support to impacted communities. Although no single economic and benefits package is capable of compensating for lost land entirely, management must nevertheless commit time and resources to improving socioeconomic quality-of-life in communities by tailoring compensation to address local needs. In recent years, a number of mining companies have provided insufficient compensation to communities, have not assumed responsibility for unanticipated events, and or have broken a series of promises. In Soroako, Indonesia, for example, INCO, one of the world’s largest nickel mining companies, began prospecting for ore in 1968, and by 1974, had fully developed a mine and an accompanying smelting complex. Although the company paid compensation to locals that relied on the land and waterways for fruit farming and drinking water, tension began shortly after operations commenced, when people complained about lowered water levels in the local lake and its feeding rivers. Although it was proven that the tailings dam was causing the disruption, INCO nonetheless blamed the indigenous peoples for the receding water levels. Moreover, despite local complaints about dust and ash emissions causing increased cases of cancer, INCO refused to commission any studies to ascertain the environmental effects of gaseous mine pollutants. The company also only hired 140 locals (drawn from 486 families and a pool of 2549 individuals) despite issuing a mandate to hire locally. Management at the Omai Mine in Guyana has also provided inappropriate compensatory packages to local communities, and has also failed to assume responsibility for unanticipated events. In August of 1995, 3.2 billion litres of cyanide-contaminated water spilled into the Essequibo River, causing damages to livestock, significant poisoning of land, and several illnesses and deaths amongst local populations. Despite being identified by many as “the worst gold mine disaster in history”, with the exception of selected

individuals, who have been awarded a mere US\$100 in settlement, Omai has not paid any compensation to locals (MiningWatch, 2000). It is therefore recommended that mine–community compensatory programmes feature, at minimum, the following:

- (1) A fully researched report providing details (issued to community groups) of the company's compensation procedures and policies, relevant stakeholder parties, and, if applicable, details of community relocation.
- (2) Direct compensation to individuals for any natural resources (e.g. trees, soils, water, etc.) and economic resources (e.g. crops, fertilizers, etc.) potentially impacted by mining activities.
- (3) In the event of relocation, provision for housing, wells, medical facilities and road infrastructure.
- (4) Direct funding for re-skilling and employment programmes for residents.
- (5) Issuance of a mandate to hire locally, and a commitment to training.

Mining companies should also contribute to local pension funds, utilize local services, and provide scholarship funding to resident universities.

A final recommendation is for mines to forge partnerships with local small-scale miners. As was indicated earlier, quite often, in a number of developing countries, small-scale miners prospect and mine the same plots of land allocated to large-scale mining operations. This principally results from a combination of research inadequacies, manpower shortages, and carelessness on the part of governments. Mine management is thus forced to endure the activity of small-scale miners, many of whom are local residents, and if they approach the issue improperly, considerable conflict can result. It is suggested here that mine management first establish cohabitation with small miners, which, as Jennings (1994) explains, is simply a voluntary agreement by which the company accepts small mining operations but specifies conditions under which they can continue operation within their concessions. Recently, a great number of large-scale mines have concerned themselves with the small-scale mining sector (UN, 1996). For example, in the Baguio Region of the Philippines, small miners commonly work within the lease boundaries of large mines and have full support of their management, who note that they “are doing a useful job” of proving portions of the ore body (Stewart, 1997). Once an agreement has been forged, management of large-scale mines can then stipulate the *specifics* of the agreement. This involves setting aside designated areas for small miners to operate, and specifying what equipment can be used. In fact, the areas that are less favourable to mine on a large-scale—more specifically, where it is unsuitable to use bulky equipment—are

usually more suitable for small-scale mining, a primary example being the reprocessing of tailings and waste dumps, which is usually only economically viable on a small-scale. Additionally, management can establish a programme for training, or involve small miners in existing training activities, where they can be educated about the potential environmental impacts of their current practices. The model example of a successful large- and small-scale mining partnership is that of AGL, Ghana. In November 1996, mine management forged a working partnership with the many illegal gold miners operating on their plot. As Appiah (1998) explains, the company first identified, registered, and regrouped the miners, and then equipped them with AGL identification cards. Further, the company banned the use of explosives, and only permitted the use of basic hand tools for excavation purposes. The gold extracted and processed by the small-scale miners is sold to a commissioned Precious Minerals Marketing Corporation (PMMC) agent located onsite. In short, the case of Abosso demonstrates that plausible working partnerships between large- and small-scale miners can be achieved.

In summary, improved (community) consultation techniques, an expanded input from regional governments, forged partnerships with small-scale miners, and the provision of adequate compensation packages to indigenous peoples are keys to minimizing land use disputes between mines and local communities. As has been outlined in this paper, land use conflicts occur in mining communities typically because of mine management neglecting the needs of indigenous people who have cultural attachments to their environment. Mines demand a significant amount of land space and in the process can prevent rural inhabitants from engaging in subsistence activities. Furthermore, as was shown in the Ok Tedi case, poor communications with locals and environmental contamination of land—the product of careless environmental management—can further intensify conflicts between mine management and surrounding communities. However, in the case of Ok Tedi and related situations, the strategies identified in this paper could prove useful in resolving and preventing a great number of land use conflicts, and help improve mining–community relations worldwide.

### Acknowledgements

This paper has only briefly examined the land use conflict issue in mining communities, and although it has focused much upon the Ok Tedi case, it is important to clarify that a great number of other communities around the world continue to dispute with the management of resident mines. The area is clearly in great need



of further research, both from environmental management and policy perspectives. The author would like to thank Professor Alexander Mather and three anonymous reviewers for their comments on earlier drafts of this article.

## References

- Aguilar, G., 2001. Access to genetic resources and protection of traditional knowledge in the territories of indigenous peoples. *Environmental Science and Policy* 4, 241–256.
- Agyapong, E., 1998. Streamlining artisanal gold mining activities and the promotion of cleaner production in the mining sector in Sub Saharan Africa: Ghana as a case study. Thesis for Master of Science in Environmental Management and Policy. International Institute for Industrial Environmental Economics, Lund University, Sweden.
- Appiah, H., 1998. Organization of small scale mining activities in Ghana. *The Journal of the South African Institute of Mining and Metallurgy* 98 (7), 307–310.
- Asp, P.J., 2000. Mining and indigenous peoples (special issue). UNEP's Industry and Environment 63.
- Castro, A.P., Nielsen, E., 2001. Indigenous people and co-management: implications for conflict management. *Environmental Science and Policy* 4, 229–239.
- Dorian, J., 1996. Minerals and mining in the transitional economies. *Mining Journal* 327 (8398), 19.
- EPA, 1995. Community consultation and involvement. Australian Environmental Protection Agency (EPA), from "Best Practice Environmental Management in Mining" series, Commonwealth of Australia.
- Epps, J., Brett, A., 2000. Engaging stakeholders. In: Otto, J.M., Cordes, J. (Eds.), *Sustainable Development and the Future of Mineral Investment*. UNEP, France (Chapter 1).
- Harper, A., Israel, A., 1999. The killing of the fly: state-corporate victimization in Papua New Guinea. Asia Pacific Working Paper 1999/22. ANU, Australia.
- Hettler, J., Lehmann, B., 1995. Environmental Impact of Large-scale Mining in Papua New Guinea: Mining Residue Disposal by the Ok Tedi Copper-gold Mine. UNEP Publications, Germany.
- Hilson, G., 2000a. Sustainable development policies in Canada's mining sector: an overview of government and industry efforts. *Environmental Science and Policy* 3, 201–211.
- Hilson, G., 2000b. Barriers to implementing cleaner technologies and cleaner production (CP) practices in the mining industry: a case study of the Americas. *Minerals Engineering* 13 (7), 699–717.
- Hilson, G., 2001a. Small-scale mining and its socioeconomic impact in developing countries. *Natural Resources Forum*, in press.
- Hilson, G., 2001b. Putting theory into practice: how has the gold mining industry interpreted the concept of sustainable development? *Mineral Resources Engineering* 10 (4), in press.
- Hilson, G., Murck, B., 2000. Sustainable development in the mining industry: clarifying the corporate perspective. *Resources Policy* 26 (4), 227–238.
- Hyndaman, D., 1997. Changing relations of production in the creation of the Ok Tedi mining enclave in Papua New Guinea. Working Paper 1997/11, Resource Management in Asia-Pacific Working Paper Series. Division of Asian and Pacific History, Australian National University.
- ILO, 1999. Social and labour issues in small-scale mines. Report for discussion at the Tripartite Meeting on Social and Labour Issues in Small-scale Mines, International Labour Organization. Sectoral Activities Programme, International Labour Office, Geneva.
- Jennings, N.S., 1994. Small-scale mining: a labour and social perspective. In: Ghose, A.J. (Ed.), *Small-scale Mining: A Global Overview*. Balkema, Rotterdam, pp. 11–18.
- King, D., 1997. The big polluter and the constructing of Ok Tedi: eco-imperialism and underdevelopment along the Ok Tedi and Fly Rivers of Papua New Guinea. In: Banks, G., Ballard, C. (Eds.), *The Ok Tedi Settlement: Issues, Outcomes and Implications*. Research School of Pacific and Asian Studies, Australian National University, pp. 94–112.
- Lemieux, A., 1997. Canada's global mining presence. In: *Canadian Minerals Yearbook*. Natural Resources Canada, Ottawa.
- Low, N., Gleeson, B., 1998. Situating justice in the environment: the case of BHP at the Ok Tedi copper mine. *Antipode* 30 (3), 201–226.
- Mbendi, 1999. Information for Africa—African Mining Statistics. Africa's leading business website. <http://www.mbendi.co.za/indy/ming/mingaf.htm#Overview>.
- McKillop, J., Brown, A.L., 1999. Linking project appraisal and development: the performance of EIA in large-scale mining projects. *Journal of Environmental Assessment Policy and Management* 1 (4), 407–428.
- McLeod, H., 2000. Compensation for landowners affected by mineral development: the Fijian experience. *Resources Policy* 26, 115–125.
- MiningWatch, 2000. On the Ground Research: A Workshop to Identify the Research Needs of Communities Affected by Large-Scale Mining. MiningWatch Canada and the Canadian Consortium for International Social Development (CCISD), Ottawa.
- Ochieng Odhiambo, M., 2000. Oxfam Karamoja Conflict Study: a Report. Oxfam, Kampala.
- Pedlowski, M.A., Dale, V.H., Matricardi, E.A.T., da Silva Filho, E.P., 1997. Patterns and impacts of deforestation in Rondonia, Brazil. *Landscape and Urban Planning* 38, 149–157.
- SOPAC, 2001. Website of The South Pacific Applied Geoscience Commission (SPAGC). <http://www.spagc.org.fj>.
- Stewart, D., 1997. Prospects and problems: small-scale gold mining in Papua New Guinea and the Philippines. In: Ghose, A.K. (Ed.), *Mining on a Small and Medium Scale*. Intermediate Technology Publications, UK, pp. 173–184.
- Steyn, M., Kahle, H., 1998. An update on the relocation of local communities at the Tarkwa Mine. In: *Mining in Africa '98*. Institute of Mining and Metallurgy, Johannesburg, pp. 15–18.
- Swales, S., Storey, A.W., Bakowa, K.A., 2000. Temporal and spatial variations in fish catches in the Fly River system in Papua New Guinea and the possible effects of the Ok Tedi copper mine. *Environmental Biology of Fishes* 57, 75–95.
- Sweeting, A.R., Clark, A.P., 2000. *Lightening the Lode: a Guide to Responsible Large-Scale Mining*. Business and Policy Group, Conservation International, Washington.
- UN, 1996. Recent developments in small-scale mining: a report of the Secretary-General of the United Nations. *Natural Resources Forum* 20 (3), 215–225.
- Vital Statistics: Mining and Canada, 1997. Drillbits and Tailings, November 8.
- Verheye, W.H., 1997. Land use planning and national soils policies. *Agricultural Systems* 53, 161–174.
- World Bank, 1998. *African Development Indicators 1998/99*. World Bank, Washington.